Sparse Sensor Placement for Dissolved Oxygen Monitoring in the Gulf of Mexico Using Genetic Algorithms

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I. Introduction

Dissolved oxygen (DO) is a critical water-quality parameter in coastal and marine ecosystems. Adequate DO concentrations are necessary to support aquatic life, and hypoxia, or a severe depletion of DO, can cause widespread fish kills, alter food webs, and damage fisheries and local economies [1]. Within the Gulf of Mexico is one of the world's largest and most studied seasonal hypoxic zones which forms annually in the northern Gulf as a result of nutrient runoff from the Mississippi River Basin [2]. Monitoring and modeling the this hypoxic zone requires long-term in-situ observations, yet the cost, logistics, and maintenance of physical sensor networks mean that complete coverage across the hundreds of thousands of square kilometers comprising the Gulf's coastal shelf is not feasible. This motivates the question, given a limited budget of physical sensors, which spatial locations should have deployed monitoring instruments so that measurements from those locations most accurately capture the broader DO field?

This project addresses the sparse sensor placement problem for DO reconstruction in the Gulf of Mexico. The objective is to determine, from a candidate set of N spatial grid cells, which subset of p sensors minimizes reconstruction error when the remaining $N - p$ cells must be estimated from those measurements alone. This is a discrete combinatorial optimization task. The number of possible sensor subsets of size p drawn from N candidates is given by the binomial coefficient:

$$\binom{N}{p} = \frac{N!}{p!\,(N-p)!} \tag{1}$$

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which grows rapidly and becomes computationally intractable even for modestly sized problems. For example, at a spatial resolution of 15 km with $N = 80$ candidate cells and a budget of $p = 40$, the number of candidate subsets exceeds 10^{22}, making an exhaustive search infeasible. This computational challenge motivates the use of heuristic search methods, and in this project a genetic algorithm (GA) is used to explore the subset space.

The relevance of this problem scopes beyond the Gulf of Mexico. In-situ oceanographic monitoring is expensive worldwide, and sparse and irregular observation networks are typical in coastal water-quality management [3]. A method for selecting which locations to monitor that accounts for the spatial variance of DO concentrations and optimizes the ability to reconstruct observed locations would have broad utility for environmental management and policy. The Gulf of Mexico dissolved oxygen data set, published by the Gulf Coast Ocean Observing System (GCOOS) [4], provides a realistic foundation for this study, with observations spanning multiple decades in a somewhat inconsistent monitoring region.

The sensor placement problem has been studied in many scientific and engineering contexts, and several algorithmic strategies have been used. Zhang et al. [5] present a sparse sensor placement strategy for sea surface temperature monitoring that frames the problem as a subset selection task over a large grid of candidate locations, which closely parallels the problem definition here. Their approach uses a greedy selection procedure in which sensor sites are chosen iteratively to maximize both reconstruction accuracy and the informational content of the sensor set. In this project, a GA is used in place of a greedy selection, motivated by the schema theorem [6], which states that short, high-fitness patterns oc-

curring in individuals in a population will propagate and combine across generations. In this context, the schema theorem will highlight sensors that are frequently chosen by the algorithm for monitoring.

Genetic algorithms are a defendable approach to structural and sensor placement optimization. Jung et al. apply a GA to the problem of selecting sensor locations for modal identification and demonstrate that the GA reliably finds configurations that outperform random and heuristic baselines [7]. In that work, the authors encode candidate sensor placements as binary strings and evolve populations using tournament selection, crossover, and mutation. This success motivated a similar strategy for this work.

The present project differs from these prior works because the underlying data is not from a clean, complete, or regularly sampled gridded field, but rather from a real-world observational network with substantial missing data and irregular temporal coverage. The GCOOS dissolved oxygen dataset consists of point observations from fixed monitoring stations and ships, oftentimes recorded at irregular intervals [4]. Converting and filtering these observations into a consistent timeline of spatial grids requires significant preprocessing, and the resulting matrix retains substantial missingness even after cleaning and spatial aggregation. This imperfect data structure creates practical challenges that makes the problem more representative of the conditions under which sensor placement decisions are actually made in practice. Handling these challenges, including imputing missing training-period values, designing an evaluation protocol for incomplete test-period coverage, and selecting an appropriate spatial resolution, accounts for a significant portion of the project and is described further in the Methods section.

This paper reports on the full pipeline from raw data to optimized sensor placements, evaluated at two spatial resolutions (15 km and 25 km grid cells) and six sensor budget levels (30% to 80% of cells). The primary metric is root mean squared error (RMSE) of the DO field reconstruction at held-out test months, measured in mg/L. The GA is compared against a random placement baseline, and results are analyzed across budget levels and spatial resolutions to determine how the quality of optimized sensor configurations changes with the number of available sensors.

II. Methodology

A. Data Source and Preprocessing

The data used in this project was obtained from the Gulf Coast Ocean Observing System (GCOOS) dissolved oxygen monitoring dataset [4], which aggregates in-situ water quality measurements from a network of fixed monitoring stations and mobile platforms across the Gulf of Mexico. The raw dataset is provided as a CSV file containing records with fields including station identifier, latitude, longitude, projected x/y coordinates, measurement timestamp, depth, and dissolved oxygen (DO) concentration in mg/L.

Initial preprocessing was performed using DuckDB [8], a high-performance database engine, as the dataset was too large to manipulate efficiently in memory using pandas. A cleaning pipeline filtered records with null required fields (DO concentration, latitude, longitude, projected coordinates, depth, and timestamp), removed physically implausible values by restricting DO concentrations to the range 0.0–25.0 mg/L, and excluded measurements at depths greater than 2 meters to focus on surface dynamics. Because the dataset includes both fixed stations and mobile platforms (e.g., vessel-based measurements), location variability was used as a proxy for platform type. Stations with more than one distinct recorded location were excluded, as they do not represent stable spatial points suitable for sensor placement. This resulted in a cleaned dataset of surface, fixed-station DO observations stored in DuckDB.

To guide the selection of a suitable study region and temporal window, an exploratory data discovery process was conducted on the cleaned dataset. Although the dataset contains over 4 million surface observations spanning 1922–2014, spatial and temporal coverage is highly uneven, with many stations reporting intermittently. Candidate regions and time windows were evaluated based on the number of observed spatial cells per month and the consistency of coverage across months. A subregion in the northern Gulf of Mexico was selected as it exhibited relatively stable coverage, with a median of over 110 observed spatial cells per month during candidate windows. A five-year period from November 2004 through October 2009 was chosen to maximize

temporal consistency while maintaining high spatial coverage.

The study region was therefore constrained to a rectangular bounding box (latitude 25.75°N–29.75°N, longitude 85.25°W–80.25°W), identified through exploratory analysis as the most densely and consistently monitored subregion of the Gulf within the dataset. The selected five-year window represents the 60-month period with the highest median spatial coverage in the cleaned data.

B. Spatial Aggregation and Matrix Construction

Because individual station locations vary slightly over time and the sensor placement problem requires a fixed, discrete set of candidate locations, raw point observations were aggregated onto a rectangular spatial grid. For each candidate cell side length s (either $s = 15{,}000$ m or $s = 25{,}000$ m), the projected coordinates (x, y) of each observation were mapped to a discrete cell index:

$$c_x = \left\lfloor \frac{x}{s} \right\rfloor, \qquad c_y = \left\lfloor \frac{y}{s} \right\rfloor \tag{2}$$

Within each (cell, month) pair, the mean DO concentration was computed across all qualifying observations. This aggregation was executed via SQL in DuckDB and the resulting cell-by-month summaries were exported to Parquet files for later use.

These Parquet files were then loaded and pivoted in Python to produce a month $\times$ cell matrix, where each entry is the mean DO concentration in a given cell during a given month. Cells with observations in fewer than 24 of the 60 months were discarded to ensure that retained cells had adequate temporal coverage for model training. This filtering yielded a 60×81 matrix at 15 km resolution and a 60×44 matrix at 25 km resolution. Overall missing data after filtering was approximately 14% at 15 km and 12% at 25 km.

C. Train/Test Split and Imputation

The 60-month matrix was partitioned into a training period (the first 48 months, covering November 2004 through October 2008) and a held-out test period (the final 12 months, covering November 2008 through October 2009). All model training, imputation, and hyperparameter choices were made exclusively on training data, with test data used only for final evaluation.

To enable Ridge regression models to be trained on complete feature vectors, the missing values in the training matrix were filled using an iterative SVD imputation procedure based on the DINEOF (Data Interpolating Empirical Orthogonal Functions) algorithm which is designed specifically for addressing missingness in oceanographic datasets [9]. DINEOF works by alternately fitting a truncated singular value decomposition to the partially observed matrix and using the low-rank reconstruction to fill missing entries, iterating until convergence. Scikit-learn's `TruncatedSVD` [10] was used with a rank of $r = 10$ components. The algorithm was run for up to 200 iterations or until the imputed values stopped changing meaningfully between iterations, defined as a relative change of less than 10^{-4}. Imputed values were clipped to $[0.0, 25.0]$ mg/L after convergence. The test matrix was left in its raw form with missing values retained, as the evaluation protocol is designed to operate on partially observed test months.

D. Reconstruction Model

The reconstruction model treats sensor placement as a supervised feature selection task. Given a selected set of p sensor cells, a separate Ridge regression model is trained for each of the remaining $N - p$ non-sensor cells, using the sensor cells' DO readings as input features and the non-sensor cell's DO reading as the target. Ridge regression is a linear regression method that includes a regularization term to prevent overfitting given that nearby cells tend to have similar DO values. Models were trained using scikit-learn [10] with regularization parameter $\alpha = 1.0$, and all models were trained on the fully imputed training matrix.

At test time, sensor readings for a given month are taken from the unimputed test matrix. A test month is included in the evaluation only if at least 80% of the selected sensor cells have non-missing observations in that month. This threshold was introduced to handle the real-world scenario in which some sensors are temporarily offline, while still ensuring that the sensor feature vector is sufficiently complete for reconstruction. For months that pass this threshold, any missing sensor values are filled with their training-period column means before prediction. Predictions are then generated for all non-sensor cells that have observed ground-truth values

in that month, and reconstruction error is measured as RMSE in mg/L pooled over all valid (month, cell) pairs in the test period. Because reconstruction error is computed over all non-sensor cells, the objective implicitly discourages spatial clustering of sensors, as over-concentration in one region leads to poor reconstruction performance elsewhere.

E. Random Baseline

Before implementing the GA, a random placement baseline was established to quantify the reconstruction performance achievable without optimization. For each combination of spatial resolution and sensor budget level (30%, 40%, 50%, 60%, 70%, and 80% of total cells), 50 independent random subsets of the appropriate size were drawn uniformly at random from the candidate cells. Each subset was evaluated using the same procedure described above, and the mean and standard deviation of RMSE across the 50 trials were recorded as the baseline for that configuration.

F. Genetic Algorithm for Sensor Placement

A GA was implemented to search for sensor configurations that minimize reconstruction RMSE. Each candidate configuration is represented as a binary string of length N, where a value of 1 at position j indicates that cell j is selected as a sensor. The budget constraint that exactly p cells must be selected is enforced after every crossover operation via a repair function that randomly removes excess selected cells or adds randomly chosen unselected cells as needed.

The population was initialized with 100 individuals, of which 50% were seeded from the cells with the highest temporal variance in the training period. The motivation is that high-variance cells carry more information about the state of the DO field, so initializing part of the population with these cells was expected to speed up early convergence. The remaining 50% of the initial population was drawn uniformly.

Selection was performed via tournament selection with a pool size of three individuals drawn at random from the population, and the individual with the lowest reconstruction RMSE was selected as a parent. Pairs of parents were combined via uniform crossover with probability 0.85, where each position in the offspring is inherited from one parent or the other with equal probability, followed by a repair step if needed. Swap mutation was applied to each offspring independently with probability 0.15, randomly exchanging one selected cell for one unselected cell. The top 5 individuals were carried forward unchanged to the next generation to prevent the loss of the best solutions found so far.

The algorithm was run for a maximum of 200 generations, with early stopping triggered if no improvement was found for 40 consecutive generations. A cache of previously evaluated configurations was maintained so that sensor subsets are never evaluated more than once. The GA was run independently for each of the 12 resolution and budget combinations, and a summary of all hyperparameters is given in Table 1. After preliminary runs, the population size was increased to 150 and patience to 60 to allow for better space exploration in lower budget levels. To assess the reliability of the algorithm, each of the twelve configurations was run 5 times independently, with a different random seed on each run, yielding 60 total GA runs. Results across runs are reported in the Stability section.

TABLE I
GENETIC ALGORITHM HYPERPARAMETERS

Parameter	**Value**
Population size	150
Informed fraction	0.50
Elite size	5
Crossover rate	0.85
Mutation rate	0.15
Tournament size	3
Max generations	200
Patience	60
Ridge α	1.0
SVD rank	10
Min obs. fraction	0.80

III. RESULTS

A. GA Performance vs. Random Baseline

Across all twelve configurations (two spatial resolutions $\times$ six budget levels), the GA-optimized sensor placements outperformed the random baseline for every test run. At 15 km resolution, the GA achieved RMSE reductions of 33% to 51% over the random baseline across budget levels of 30–80%, with the largest gains occurring at higher budgets. At 25 km resolution, improvements ranged from 38% to 56%, again with the most pronounced gains

Algorithm 1 Genetic Algorithm for Sensor Placement

```
Initialize population P_0 of size 150 (50% informed, 50% random)
Evaluate fitness for all individuals in P_0
for generation g = 1 to 200 do
    Carry elite (e = 5 best) into P_g
    while |P_g| < 150 do
        Select parents a, b via tournament (k = 3)
        if rand() < 0.85 then
            Apply uniform crossover → offspring c_1, c_2; repair
        else
            c_1 ← a, c_2 ← b
        end if
        if rand() < 0.15 then
            Apply swap mutation to c_1
        end if
        if rand() < 0.15 then
            Apply swap mutation to c_2
        end if
        Add c_1, c_2 to P_g
    end while
    Evaluate fitness for new individuals (cache hits skip re-evaluation)
    if no improvement for 60 generations then
        break
    end if
end for
return best individual found
```

at larger sensor budgets. In terms of absolute performance, GA-optimized RMSE at 15 km decreased from 1.052 mg/L at the 30% budget to 0.787 mg/L at the 80% budget, while at 25 km resolution RMSE decreased from 0.896 mg/L to 0.697 mg/L over the same range.

The random baseline RMSE remained approximately flat as budget increased at both resolutions, indicating that randomly selected sensors provide limited reconstruction benefit even as more sensors are added. By contrast, the GA consistently identified configurations whose RMSE decreased with budget, suggesting that the algorithm is able to exploit additional sensors to capture spatial structure in the DO field rather than redundantly sampling similar regions.

These trends are visualized in Figure 1, where the GA curve exhibits a clear downward trajectory with increasing budget, while the random baseline remains nearly constant. Figure 2 further shows that the GA reliably improves solution quality over generations across all configurations, with rapid initial decreases in RMSE followed by convergence to stable plateaus. Notably, the gap between mean and best RMSE narrows over time, indicating consistent population-wide improvement rather than reliance on a small number of high-performing individuals.

To assess the impact of population initialization, convergence behavior was also evaluated without informed pre-seeding (Figure 2a) and with 50% variance-based pre-seeding (Figure 2b). Pre-seeding leads to faster early-stage convergence, particularly at lower sensor budgets, as initial individuals are closer to high-quality regions of the search space. However, both approaches converge to nearly identical final RMSE values across all configurations, indicating that pre-seeding improves convergence efficiency but does not affect solution quality.

Figure 4 shows the distribution of per-month reconstruction error across the 12 test months for each budget level. At lower budgets, the spread of RMSE values is wider, indicating greater variability in reconstruction quality across months. As the budget increases, both the median RMSE and the interquartile range decrease, suggesting that higher sensor counts not only improve average performance but also yield more consistent reconstructions across varying temporal conditions. All figures were produced using Matplotlib [11].

B. Stability Across Independent Runs

To assess the reliability of the GA, each of the twelve configurations was run 5 times independently, with a different seed randomly chosen using `np.random.default_rng()`, yielding 60 total GA runs. The best RMSE achieved on each independent run was recorded, and the mean and standard deviation were computed per configuration, reported in Table 2.

Overall, the GA demonstrated strong stability across independent runs. Standard deviations in best RMSE were small across all configurations, ranging from 0.0025 to 0.0280 mg/L. The largest variability was observed at the 15 km resolution, 60% budget configuration ($\sigma = 0.0280$ mg/L), suggesting that at this setting multiple distinct sensor subsets

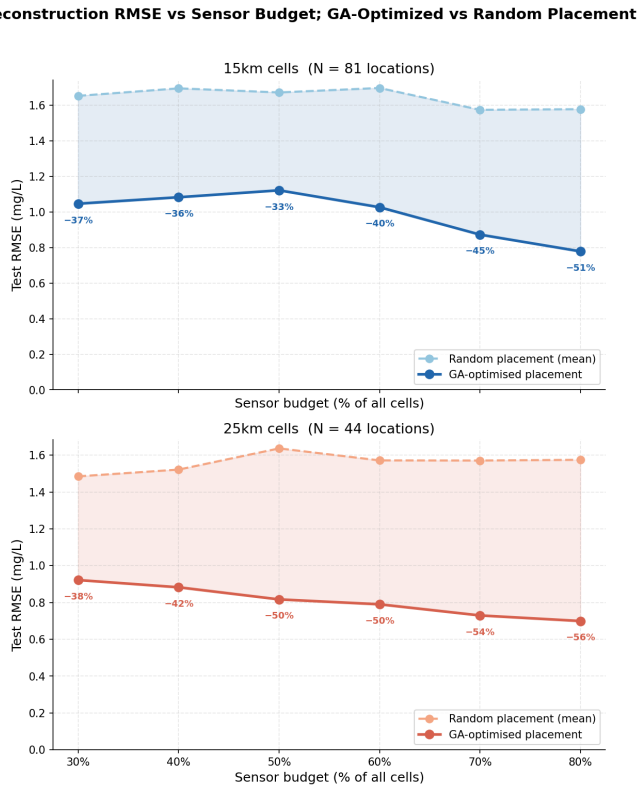

Fig. 1. Reconstruction RMSE vs. sensor budget for GA-optimized and random placement at 15 km and 25 km resolution. Percentage labels indicate RMSE reduction relative to the random baseline.

can achieve similar reconstruction error and the algorithm does not always converge to the same solution. The most stable configuration was the 25 km, 80% budget, which had a standard deviation of just 0.0025 mg/L, indicating that the algorithm consistently converged to nearly identical solutions regardless of initialization.

C. Sensitivity to Spatial Resolution

Comparing results across the two spatial resolutions reveals a consistent pattern of lower absolute RMSE at lower resolution. At 25 km, the grid achieves lower absolute RMSE than the 15 km grid at every budget level, despite having fewer candidate cells ($N = 44$ vs. $N = 81$). At the 80% budget, for example, the best GA RMSE was 0.697 mg/L at 25 km compared to 0.787 mg/L at 15 km. This likely reflects the fact that aggregating observations into larger cells smooths local noise or variability and reduces the complexity of the spatial field, making it more suitable to linear reconstruction from a small sensor set. However, this comes at the cost of spatial detail, and the 15 km results may be more meaningful for applications requiring fine-grained mapping of hypoxic boundaries.

D. Sensor Placement Patterns

Figure 3 shows the GA-selected sensor locations at 30% and 70% budget for both spatial resolutions, overlaid on the training-period mean DO concentration. At the 30% budget, selected sensors at both resolutions are concentrated in the upper-left portion of the study region, corresponding to the cells with the highest mean DO values, as well as along the central diagonal spine of the monitoring network. The lower-right portion of the region, where mean DO is lower, receives comparatively few sensors at this budget level. As the budget increases to 70%, the selected set expands to cover nearly the full spatial extent of the grid, filling in the southern and eastern portions that were deprioritized at lower budgets. The upper-left cluster is consistently selected across both budget levels and both resolutions, suggesting that the GA identifies these high-DO boundary cells as the most informative locations for reconstructing the broader field, consistent with the finding of Zhang et al. [5] that boundary and high-variability regions tend to be prioritized by optimized placement strategies.

E. Reconstruction Consistency Across Test Months

Figure 4 shows the distribution of per-month reconstruction RMSE across the 12 held-out test months for each sensor budget level. At lower budgets, the spread of monthly RMSE values is wide, indicating that reconstruction quality varies considerably from month to month and that some months are substantially harder to reconstruct than others. As the budget increases, the inter quartile range narrows noticeably, particularly at 25 km resolution, suggesting that higher sensor counts produce more consistently accurate reconstructions regardless of the specific month being evaluated. The presence of high outliers at lower budgets at both resolutions indicates that a small number of test months remain difficult to reconstruct even with optimized placement, likely reflecting months with higher-than-average data sparsity among the selected sensors.

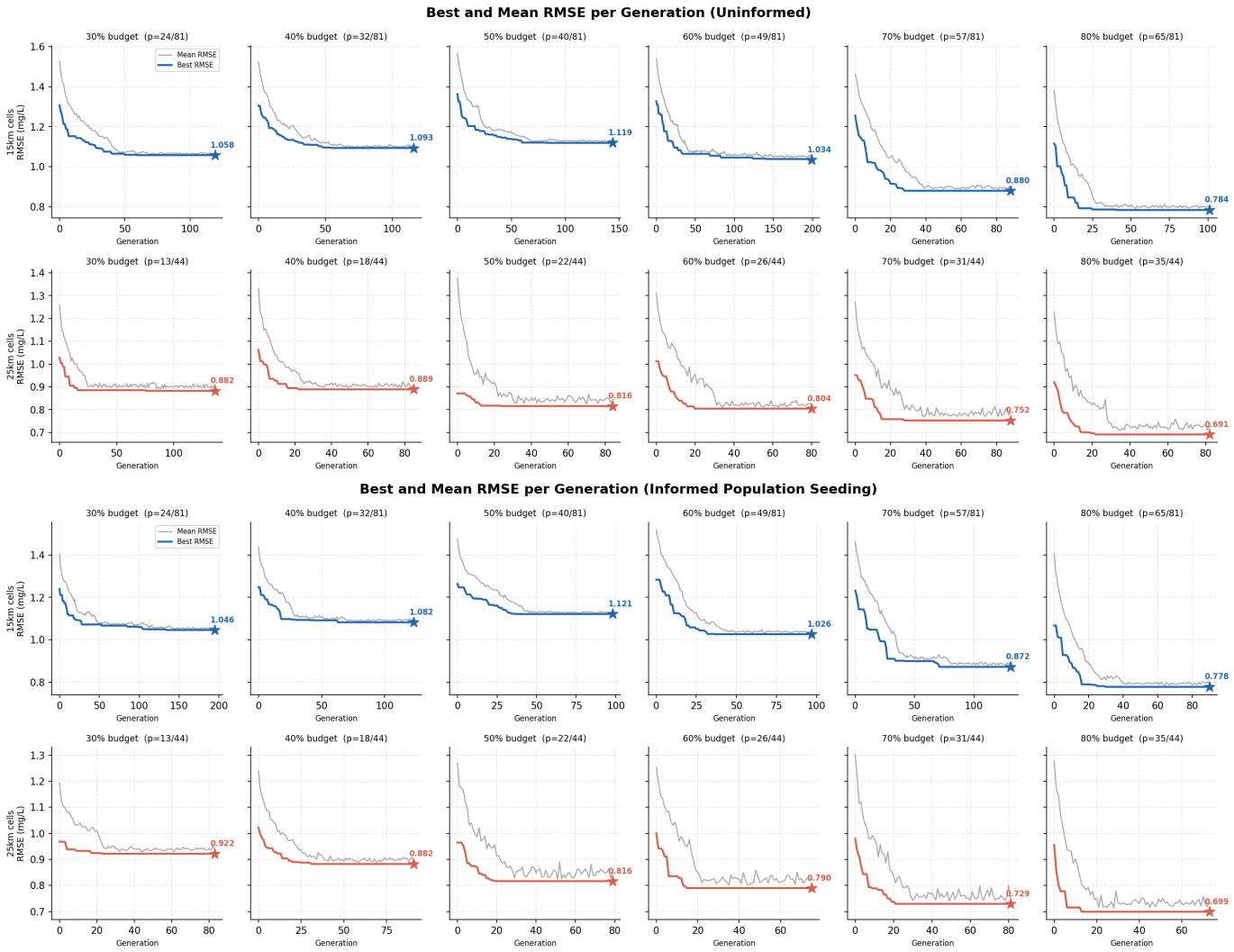

Fig. 2. GA convergence curves showing best and mean RMSE per generation across all twelve configurations. (a) Convergence without informed population seeding. (b) Convergence with 50% variance-based pre-seeding. Pre-seeding results in faster early-stage convergence, particularly at lower sensor budgets, but both approaches reach comparable final RMSE.

IV. Conclusion

This project presented a full pipeline for sparse sensor placement optimization applied to dissolved oxygen monitoring in the Gulf of Mexico, from raw data preprocessing through genetic algorithm optimization and stability evaluation. The core objective was to determine which subset of spatial grid cells, when instrumented as sensors, best enables reconstruction of the broader DO field under a fixed sensor budget. Results demonstrate that the genetic algorithm consistently and substantially outperformed random sensor placement across all twelve configurations tested, achieving RMSE reductions of 33–51% at 15 km resolution and 38–56% at 25 km resolution relative to the random baseline. Improvement increased with sensor budget at both resolutions, suggesting that the GA is particularly effective at exploiting the ability to capture spatial DO variability when more sensors can be deployed.

The stability analysis across five independent runs confirmed that these results are reliable and not a product of of a favorable random initialization. Standard deviations in best RMSE were small across all configurations, ranging from 0.0025 to 0.0280 mg/L, and the algorithm converged to consistent solutions in the large majority of cases. The one exception was the 15 km, 60% budget configuration, where slightly higher variance suggested that multiple distinct sensor subsets can achieve

TABLE II
GA STABILITY RESULTS ACROSS 5 INDEPENDENT RUNS

Resolution	Budget	p	Random Baseline RMSE	Mean GA RMSE	Std	Improv.
15km	30%	24	1.6524	1.0524	0.0081	36.3%
15km	40%	32	1.6946	1.0706	0.0069	36.8%
15km	50%	40	1.6716	1.1074	0.0083	33.8%
15km	60%	49	1.6966	1.0116	0.0280	40.4%
15km	70%	57	1.5739	0.8867	0.0106	43.7%
15km	80%	65	1.5774	0.7868	0.0074	50.1%
25km	30%	13	1.4847	0.8958	0.0162	39.7%
25km	40%	18	1.5215	0.8460	0.0188	44.4%
25km	50%	22	1.6366	0.8188	0.0052	50.0%
25km	60%	26	1.5714	0.7991	0.0071	49.1%
25km	70%	31	1.5707	0.7309	0.0132	53.5%
25km	80%	35	1.5746	0.6965	0.0025	55.8%

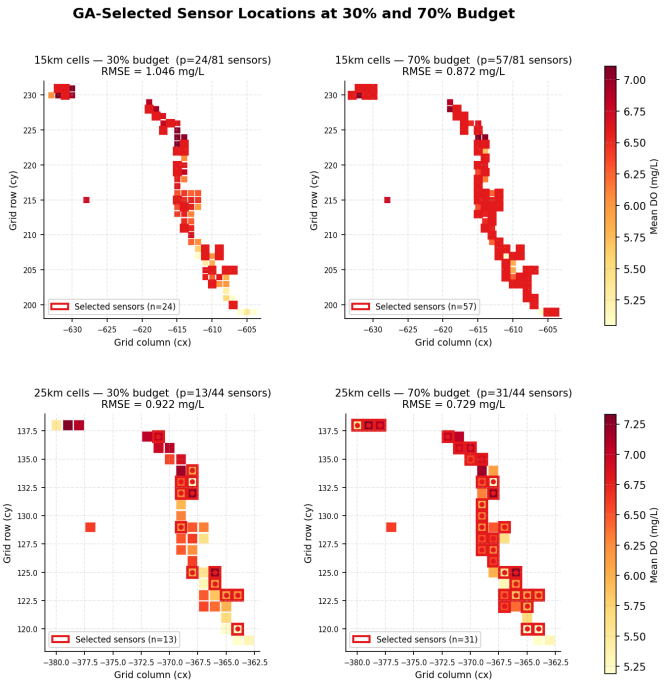

Fig. 3. GA-selected sensor locations at 30% and 70% budget for both spatial resolutions. Background colour indicates training-period mean DO concentration. Red borders mark selected sensor cells.

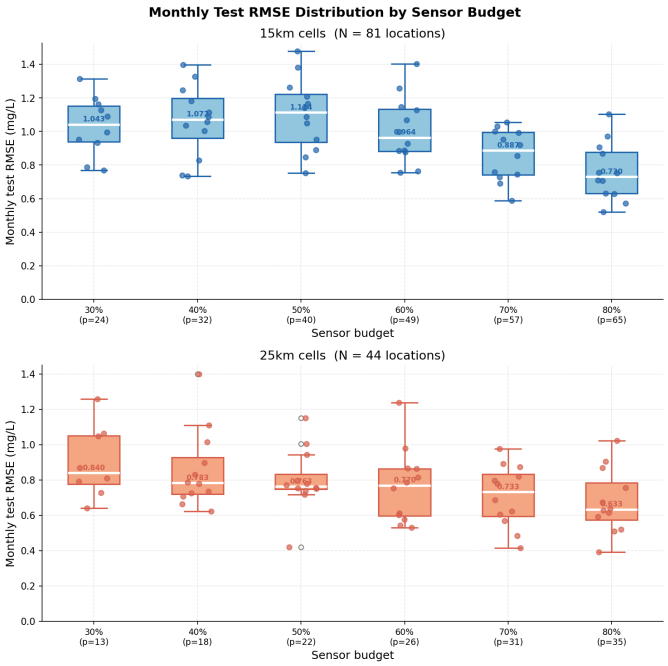

Fig. 4. Distribution of per-month reconstruction RMSE across the 12 test months for each sensor budget level. Individual points show monthly RMSE values and the white line marks the median.

similar reconstruction error at that setting.

The 25 km grid achieved lower absolute RMSE than the 15 km grid at every budget level, which likely reflects the smoothing effect of aggregating observations into larger cells rather than a fundamental advantage of coarser resolution. However, the 15 km results are arguably more scientifically meaningful for applications such as mapping the spatial extent of hypoxic zones.

This project differs from prior work such as Zhang et al. [5] in that it operates on real-world observational data with substantial missing values and irregular temporal coverage, rather than a clean gridded field. The challenges this introduced make the problem more representative of real conditions faced by environmental monitoring agencies. The results suggest that even under these imperfect data conditions, a genetic algorithm can identify sensor placements that dramatically outperform random selection, providing a quantitative basis for guiding monitoring network design in the Gulf of Mexico and potentially in other coastal observing systems

facing similar data constraints.

Future work could extend this approach in several directions. Incorporating information entropy into the fitness function, as proposed by Zhang et al. [5], could further improve placement quality by explicitly rewarding sensors in high-variability regions. Expanding the study period or geographic extent, running the optimization at finer spatial resolutions, or applying the pipeline to other water quality parameters such as temperature or chlorophyll concentration would also broaden the utility of the framework. Finally, a secondary analysis deriving hypoxic area estimates from reconstructed DO fields would provide a direct measure of the scientific value of optimized sensor placements for the primary application motivating this work.

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